

# SnapNet

HEN Network Diagrams from SnapOpt Solutions — User Guide

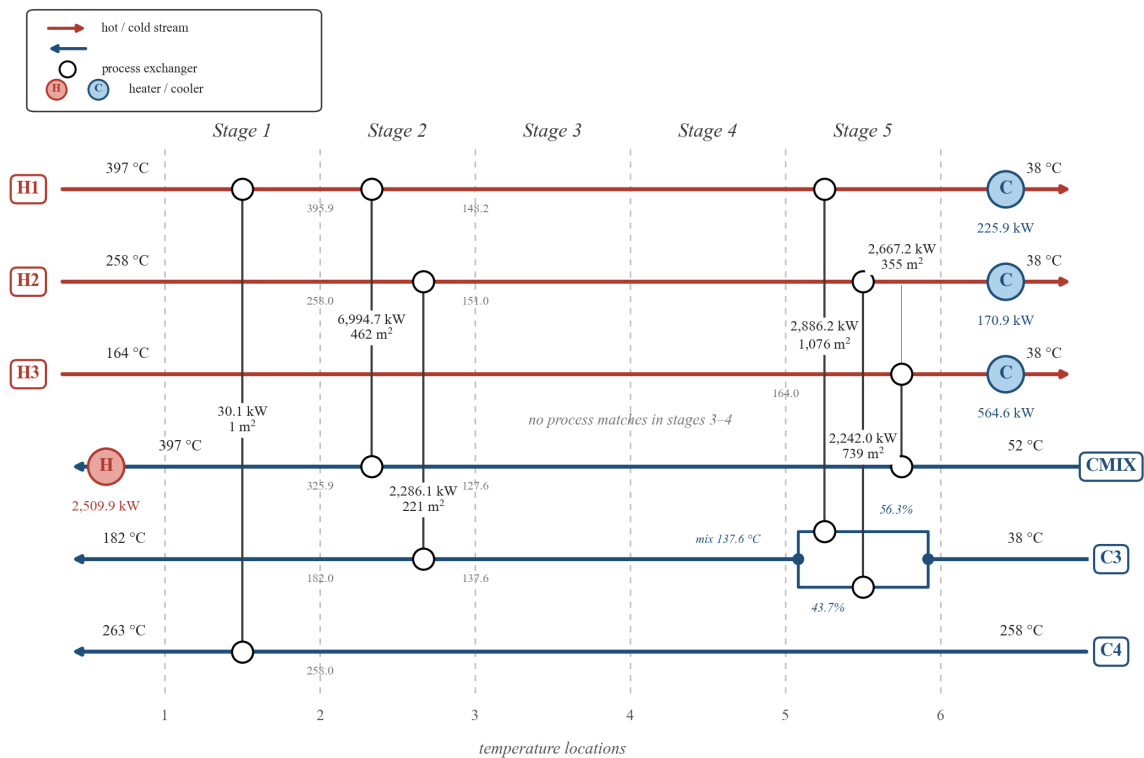
## User Guide

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SnapTera: From Targeting to Optimization to Network Design

Optimized heat exchanger network (SSGO) | TAC = \$364,077/yr  $Q_H = 2,510$  kW  $Q_C = 961$  kW 10 units



SnapNet turns a SnapOpt solution into the final deliverable: a publication-ready stagewise grid diagram of the optimized heat exchanger network — stream lines, every exchanger with its duty and area, coolers and heaters, stage-boundary temperatures, and automatic stream splits with duty fractions and mixing temperatures.

## 1. Requirements and quick start

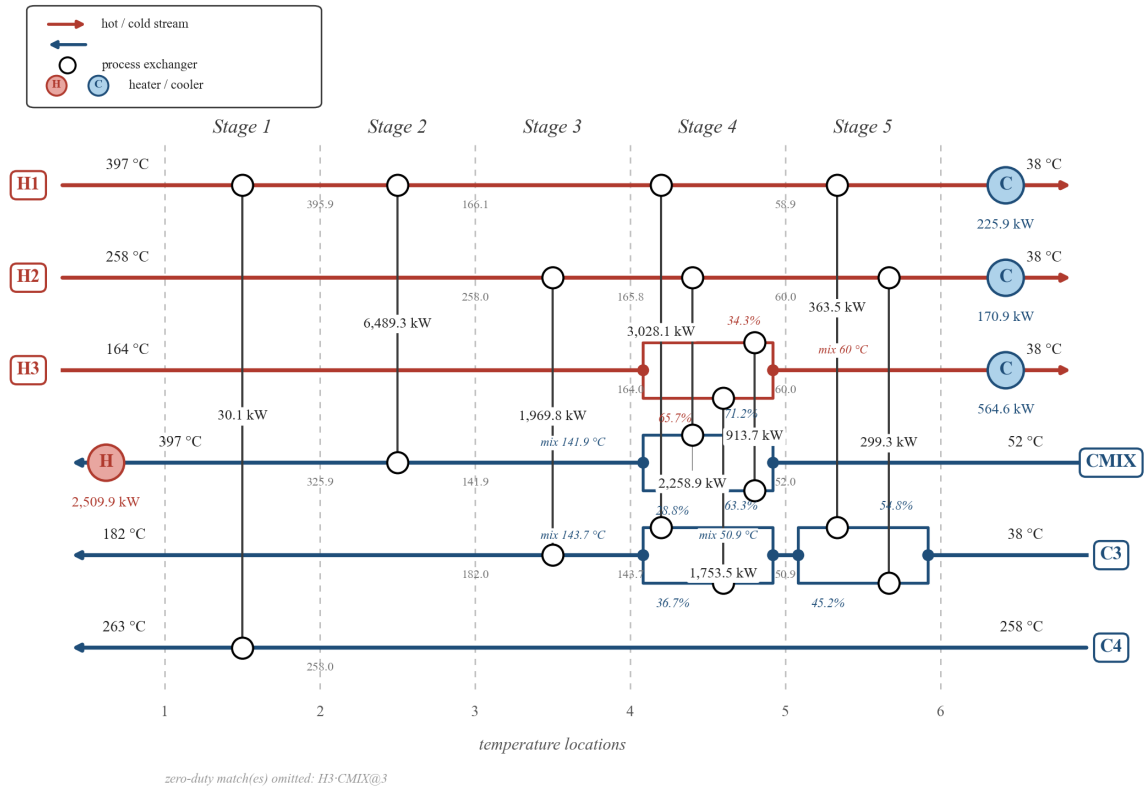
SnapNet is a single Python script (`snapnet_draw.py`, needs Python 3 with matplotlib). Its input is the `solution.json` that SnapOpt writes into every run's `results\` folder.

| Command                                                       | Effect                                                                                             |
|---------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| <code>python snapnet_draw.py</code>                           | Draw the default solution (SnapOpt\results\solution.json) to <code>hen_grid.png</code> next to it. |
| <code>python snapnet_draw.py run\results\solution.json</code> | Draw a specific SnapOpt run.                                                                       |
| <code>... -o net.png --areas</code>                           | Choose the output file; print exchanger areas under the duties.                                    |
| <code>... --svg --pdf --dpi 600</code>                        | Also export SVG and PDF; raise the PNG resolution.                                                 |
| <code>... --streams input.json</code>                         | Use exact stream data (auto-detected next to the solution when present).                           |
| <code>... --no-temps --title "My network"</code>              | Hide stage-boundary temperatures; custom title.                                                    |

## 2. Reading the diagram

| Element                   | Meaning                                                                                                                                                                                                      |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Red / blue arrows         | Hot streams flow left→right, cold streams right→left; supply and target temperatures at the line ends.                                                                                                       |
| White circles + connector | A process exchanger between the two streams, with its duty (kW) and, with <code>--areas</code> , its area (m <sup>2</sup> ). A thin gray leader line ties a label to its exchanger wherever space was tight. |
| H / C circles             | Fired heater (on cold streams, left) and cooler (on hot streams, right) with utility duties.                                                                                                                 |
| Dashed grid, Stage k      | The superstructure stages and temperature locations; small gray numbers under each line are the stream's stage-boundary temperatures.                                                                        |
| Forked line               | A stream SPLIT: parallel branches with duty fractions (%), re-mixed isothermally at the temperature shown after the merge node.                                                                              |
| Footnote                  | Zero-duty matches (solver bookkeeping artifacts) are omitted and listed.                                                                                                                                     |

Optimized heat exchanger network (SSGO) | TAC = \$375,462/yr  $Q_H = 2,510$  kW  $Q_C = 961$  kW 14 units



A dense diagnostic case (DICOPT local incumbent): four simultaneous splits in stages 4–5, including a hot-side split of H3 — drawn automatically.

### 3. Where the stream data comes from

The diagram needs each stream's CP and supply/target temperatures. SnapNet looks for the run's `input.json` next to the solution (written by every SnapOpt run) and uses those exact values. When there is none — e.g. a solution imported straight from a GAMS listing — SnapNet derives the data from the solution itself via stage energy balances: CP from the largest-duty stage (duty / temperature change), targets from the utility duties. On the thesis case the derived CPs match the true values to  $\sim 3 \cdot 10^{-4}$  relative.

### 4. Validation

The automated test (`test_snapnet_draw.py`) checks the drawn network against the hand-built thesis figure: the C3 split in stage 5 at 56.3 / 43.7 % with isothermal mixing at 137.6 °C, all six matches and duties, the derived stream table, and — on the DICOPT case — all four splits and the omitted zero-duty match.

SnapNet v1.0 — created by Mohammed Ali Mansoury, 2026. MIT License. Part of the SnapTera suite with SnapTarget (targeting) and SnapOpt (optimization). Contact: [atsnaptera@gmail.com](mailto:atsnaptera@gmail.com)